PROGRAM 7: Demonstration and Analysis of Hallucinations

AIM

To demonstrate and analyze hallucinations in a pretrained language model using reference information.

ALGORITHM

1. Import the Transformers library.
2. Load a pretrained FLAN-T5 model.
3. Define reference information.
4. Ask questions related to the reference.
5. Generate responses.
6. Compare responses with the reference.
7. Identify unsupported answers.

CODE

```
from transformers import pipeline


model = pipeline(
    "text2text-generation",
    model="google/flan-t5-small"
)


reference = """
Saveetha Engineering College is located in Chennai.
It offers engineering courses in different disciplines.
"""


questions = [
    "Where is Saveetha Engineering College located?",
    "Who is the principal of Saveetha Engineering College?"
]


for question in questions:
```

```
prompt = """
Answer only using the reference information.
If the answer is unavailable, say I don't know.

Reference:
""" + reference + """

Question: """ + question

result = model(
    prompt,
    max_new_tokens=60
)

print("\nQuestion:", question)
print("Answer:", result[0]["generated_text"])
```

INPUT

Reference:

Saveetha Engineering College is located in Chennai.

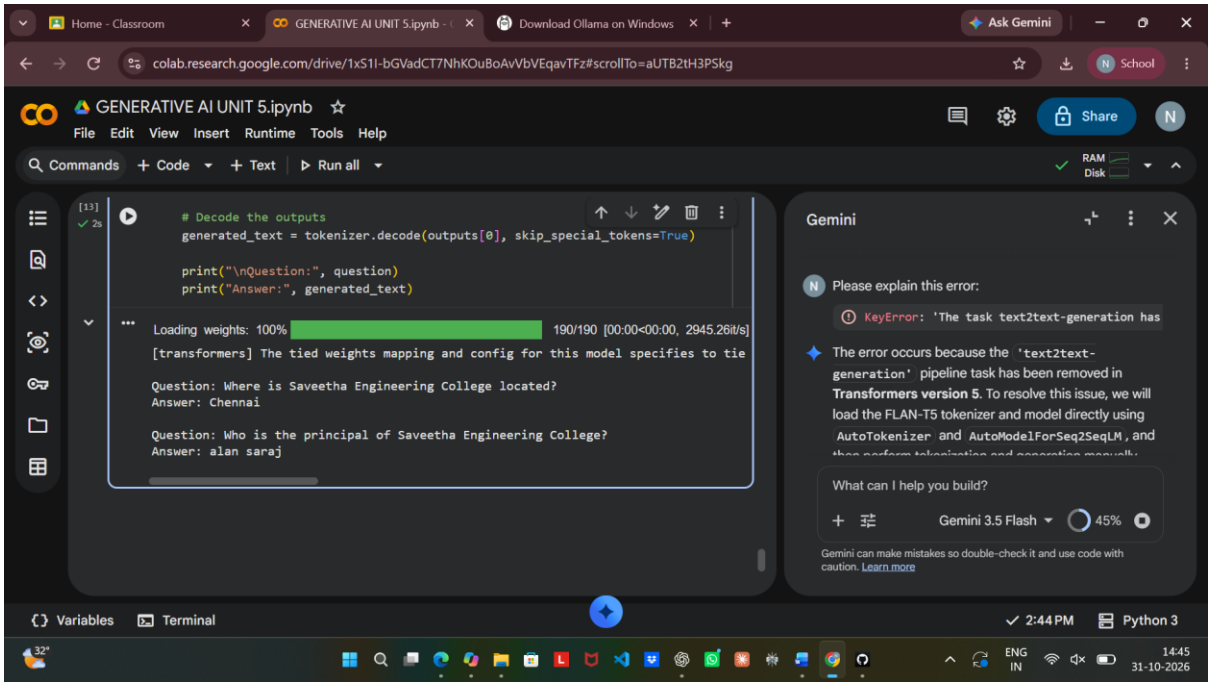
Questions:

1. Where is Saveetha Engineering College located?
2. Who is the principal of Saveetha Engineering College?

OUTPUT

The first question should produce Chennai.

The second question should ideally produce an answer indicating that the information is unavailable.



ANALYSIS

- Correct information is grounded in the reference.
- Invented information is a possible hallucination.
- Language models can generate incorrect information even when instructed to avoid guessing.

RESULT

Thus, hallucinations were demonstrated and analyzed using a pretrained language model.